

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Final Model & Data Science Package

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance
and Healthcare Support Using Data and AI
Track: Data Science

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Week 8 Submission – Final Model Selection, Documentation & Presentation

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This report is the Week 8 Data Science track deliverable for the HealthConnect Clinic Experience Lab engagement: the HealthConnect Final Model & Data Science Package. Week 8 does not repeat model selection, hyperparameter tuning, or any part of the Week 5–7 development and testing cycle. Instead, it formally confirms the Week 7 tested and refined candidate as the final model of record, consolidates the evaluation evidence already produced, and documents that model for cross-track integration, business audiences and the final HealthConnect presentation. Every figure reported below is taken directly from the executed output of the accompanying Week 8 notebook (Final_model_selection_documentation_presentation.ipynb), which itself loads the frozen Week 7 artefacts rather than regenerating them, and every figure has been verified against that executed output rather than against notebook narrative, in line with this track's established practice.

This report draws on four prior Data Science deliverables — the Week 5 HealthConnect Baseline Modelling Report, the Week 6 HealthConnect Model Improvement & Validation Report, the Week 7 HealthConnect Model Testing & Refinement Report, and the Week 7 HealthConnect Assumptions, Limitations, Risks & Dependencies Report — together with the validated Week 7 findings from the Data Analytics track and the Week 8 HealthConnect Data Analytics Assignment brief, which sets out the mandatory structure followed in Sections 2 and 12 below.

1. Week 7 → Week 8 transition

This concise transition summary carries forward the key outcomes of Week 7 without reproducing that report; the full testing evidence remains in the Week 7 Model Testing & Refinement Report and its standalone Assumptions, Limitations, Risks & Dependencies companion document.

- **Main Week 7 output:** a systematically tested and validated refined Logistic Regression candidate model for predicting appointment no-show risk, built on the full 38-feature specification.
- **Most important testing result:** the candidate did not demonstrate a large or consistently reproduced improvement over the Week 5 baseline across cross-validation folds, but it provides a transparent, operationally interpretable probability-based model suitable for final integration.
- **Major issue discovered:** testing identified variation in selected calendar-feature coefficients and distinct false-negative/false-positive patterns across appointment segments; the operational threshold also involves a material trade-off between identifying more no-shows and increasing supportive outreach to patients who would attend regardless.
- **What was refined:** a targeted calendar-feature ablation (removing `appointment_day_of_year` and `appointment_month`) was tested. The proposed simplification was rejected because validation Recall@0.45 declined by 0.0021, marginally exceeding the predefined 0.002 acceptance tolerance.
- **Component now considered ready:** the final 38-feature refined Logistic Regression specification is frozen and ready for final documentation and integration testing.
- **What still requires integration:** the final model specification, preprocessing requirements, feature schema, probability output and provisional decision threshold must be formally communicated to, and validated by, ML Engineering.

- **Tracks involved in final integration:** Data Analytics → Data Science, Data Science → ML Engineering, ML Engineering → Data Science, and Project Management for overall final integration/presentation coordination.
- **Final presentation objective:** demonstrate how validated attendance patterns, predictive modelling, and cross-track integration can support HealthConnect in prioritising supportive attendance interventions, while clearly communicating the model's limitations.

2. Final integration readiness

In line with the Week 8 brief's Part 1 requirement, the following table establishes that the Data Science component is ready for final integration before the HC-POD walkthrough and presentation.

Requirement	Response
What is my final component?	38-feature refined Logistic Regression no-show risk model.
What problem does it address?	Identification of appointments with elevated no-show risk, to support prioritised supportive administrative intervention.
What is its current status?	Model testing complete; ready for final integration validation.
What did I improve during Week 7?	Performed CV stability testing, FP/FN analysis, segment testing, feature-stability analysis, threshold validation, and a calendar-feature ablation test.
Which other tracks does it depend on?	Data Analytics (validated attendance-pattern findings) and ML Engineering (pipeline compatibility confirmation).
Which tracks depend on my output?	ML Engineering (final model and scoring contract) and Project Management (final integration and presentation coordination).
What output will I provide?	Final model artefact, 38-feature schema, preprocessing requirements, probability-output definition, threshold guidance and documented limitations.
What output do I need from another track?	Pipeline compatibility results from ML Engineering, confirming that preprocessing, feature order and model scoring are reproduced correctly.
What evidence demonstrates readiness?	Week 7 cross-validation, holdout evaluation, error analysis, ablation retest, and cross-track Data Analytics validation.
What limitations remain?	Moderate discrimination; provisional threshold; false-positive burden; non-grouped cross-validation; preprocessing not fully nested in cross-validation; integration and identifier-traceability dependencies.

3. Final candidate model confirmation

Week 8 performs no active model selection or hyperparameter tuning. The final tested Week 7 outcome is instead frozen as the candidate of record, and formally reviewed and documented here for cross-track handoff and solution integration. Confirmation relies on the serialised Week 7 model artefact, the finalised 38-feature schema, consolidated holdout evaluation

results, fold-level cross-validation stability records, the documented feature-ablation decision, and cross-track validation results from Data Analytics.

Component	Final specification
Final model	Refined Logistic Regression
Model family	LogisticRegression (scikit-learn)
Final feature count	38
Probability output	Probability of appointment No-Show
Operational threshold	0.45
Threshold status	Provisional — pending operational capacity/cost agreement
Week 7 refinement outcome	Calendar-feature ablation rejected; full specification retained
Week 8 status	Final candidate confirmed for integration

Verification confirms that the saved feature schema contains exactly the expected 38 features, with unique names and a frozen order, and that the serialised model object is a scikit-learn LogisticRegression instance expecting 38 input features. The proposed calendar-feature ablation was not adopted: mean validation Recall@0.45 fell by 0.0021, marginally exceeding the strict predefined acceptance tolerance of 0.002. This reflects systematic adherence to a predefined model-selection rule rather than a substantively large or statistically significant predictive loss, and the complete 38-feature specification is accordingly preserved and carried forward into the final HealthConnect integration suite. The 0.45 operational threshold remains provisional: it provides a working baseline for integration testing but requires formal stakeholder review to align outreach capacity, staff workload, and the relative business costs of false positives versus false negatives.

Final 38-feature schema

The final feature schema was validated to contain exactly 38 unique, correctly ordered features, combining patient, booking, appointment, and engineered interaction variables:

#	Feature
1–5	age; booking_lead_days; previous_appointments; previous_no_shows; distance_to_clinic_km
6–12	booking_month; booking_day_of_week; booking_day_of_year; booking_week_of_year; appointment_month; appointment_day_of_year; appointment_week_of_year
13–19	gender_Male; gender_Prefer not to say; age_group_25-34; age_group_35-44; age_group_45-54; age_group_55-64; age_group_65+
20–22	appointment_type_Follow-up; appointment_type_General Consultation; appointment_type_Specialist Consultation
23–28	appointment_day_Monday; appointment_day_Saturday; appointment_day_Sunday; appointment_day_Thursday; appointment_day_Tuesday; appointment_day_Wednesday
29–34	appointment_time_Evening; appointment_time_Morning; reminder_sent_Yes; reminder_channel_No Reminder; reminder_channel_SMS;

#	Feature
	reminder_channel_WhatsApp
35–38	is_sunday_appointment_1; history_x_lead_time; distance_x_lead_time; history_interaction

4. Final model evaluation

The final model is evaluated using the frozen Week 7 holdout results; no additional model selection or threshold optimisation has been performed on this holdout dataset during Week 8. The objective here is to consolidate the validated Week 7 performance metrics directly into the final Data Science package.

Metric	Final model (threshold = 0.45)
Accuracy	0.6263
Precision	0.6066
Recall	0.7184
F1	0.6578
F2	0.6929
ROC-AUC	0.6890
PR-AUC	0.6717

Final confusion matrix

Outcome	Count	Share of holdout
True Negative (TN)	258	26.71%
False Positive (FP)	225	23.29%
False Negative (FN)	136	14.08%
True Positive (TP)	347	35.92%

At the provisional 0.45 threshold, the final model achieved a holdout ROC-AUC of approximately 0.689 and a PR-AUC of approximately 0.672. It identified approximately 71.8% of observed no-shows (recall) at a precision of approximately 60.7%. Lowering the decision threshold from the conventional 0.50 default to 0.45 successfully captured more true no-shows at the cost of a higher false-positive rate; the selection of 0.45 therefore remains provisional until HealthConnect defines acceptable outreach capacity constraints and the relative business costs of false negatives versus false positives.

5. Baseline vs final candidate comparison

Cross-validation comparison between the Week 5 baseline and the Week 7/Week 8 final candidate, computed across the same five stratified folds used throughout Week 7 testing, is summarised below.

Metric	Week 5 baseline (mean)	Week 7/8 candidate (mean)	Difference
ROC-AUC	0.6756	0.6750	-0.0007
PR-AUC	0.6819	0.6812	-0.0007

Fold-by-fold improvement consistency

Metric	Folds won	Folds lost	Ties	Total folds	Win %
ROC-AUC	3	2	0	5	60.0%
PR-AUC	2	3	0	5	40.0%

The fold-level comparison shows only mixed and inconsistent improvement over the Week 5 baseline: the candidate's ROC-AUC was higher in 3 of 5 folds (60%) and its PR-AUC was higher in only 2 of 5 folds (40%), and the mean cross-validated difference on both ranking metrics is effectively zero (-0.0007). The final candidate does not demonstrate a clear, consistently reproduced improvement in predictive discrimination over the baseline. It is retained instead on the strength of the broader Week 6–7 evidence base — its interpretability, probability-based risk scoring, tested 38-feature specification, operational-threshold analysis, and integration readiness — rather than because it is a materially more accurate classifier. Accordingly, the final HealthConnect model should be presented as a moderate risk-prioritisation model, not as a highly accurate or production-ready predictive system.

6. Error analysis summary

Week 7 conducted detailed false-positive, false-negative and segment-level error analysis; Week 8 consolidates those findings to explain the final model's operational strengths and weaknesses rather than repeating the underlying testing.

Outcome	Count	% of holdout	HealthConnect interpretation
True Positive	347	35.92%	No-show correctly identified for possible supportive outreach.
True Negative	258	26.71%	Attending appointment correctly left unflagged.
False Negative	136	14.08%	No-show missed by the model — a missed opportunity for support.
False Positive	225	23.29%	Attending appointment flagged — additional outreach burden.

False negatives (136 cases, 14.08% of the holdout) represent patients who ultimately missed their appointment but were not flagged, a missed opportunity for proactive support. False positives (225 cases, 23.29% of the holdout) represent patients who would have attended regardless but were flagged, imposing an unnecessary administrative outreach and contact burden on clinic staff. Lowering the threshold from 0.50 to 0.45 captured a larger share of actual no-shows at the cost of a higher false-positive rate; Week 7's segment-level testing additionally confirmed that this error behaviour, particularly the false-positive rate, varies noticeably across appointment segments such as booking lead-time band and previous no-show history. The 0.45 threshold therefore remains provisional until HealthConnect leadership

defines acceptable operational capacities, outreach resources, and the relative business costs of false negatives versus false positives.

7. Feature and model decisions

Week 7 investigated whether removing two calendar-based features (appointment_day_of_year and appointment_month) could simplify the final model without a material reduction in predictive performance.

Metric	Mean (current, 38-feature)	Mean (ablated)	Mean change
ROC-AUC	0.674951	0.675353	+0.000402
PR-AUC	0.681217	0.682226	+0.001010
Recall@0.45	0.721649	0.719588	-0.002062
F1@0.45	0.664436	0.663674	-0.000763

The ablated model produced only negligible, minor changes in overall ranking metrics (ROC-AUC and PR-AUC), but mean Recall@0.45 deteriorated by approximately 0.0021 on the validation sample. Because this reduction marginally exceeded the strict predefined acceptance tolerance of 0.002, the proposed ablation was formally rejected. This decision reflects adherence to a predefined model-selection rule rather than a statistically significant or substantively large predictive loss, and the final Week 8 model consequently retains the complete 38-feature specification, as confirmed in Section 3.

8. Business interpretation and responsible use

Strengths and weaknesses

Area	Assessment	Evidence / meaning
Interpretability	Strength	Logistic Regression provides a comparatively transparent final model.
Risk ranking	Moderate strength	Holdout ROC-AUC is approximately 0.689, indicating useful but moderate discrimination.
No-show identification	Strength with trade-off	Recall at 0.45 is approximately 0.718, but higher recall comes with additional false positives.
Probability output	Strength	The model produces probabilities that support flexible risk prioritisation.
Model improvement	Limitation	Cross-validation shows mixed baseline improvement rather than a consistently superior model.
False positives	Limitation	225 attending appointments were flagged at the 0.45 holdout threshold.
Segment performance	Limitation requiring monitoring	Week 7 identified variation in FP/FN behaviour across appointment segments.
Operational	Unresolved	0.45 remains provisional pending outreach-capacity and

Area	Assessment	Evidence / meaning
threshold	operational decision	cost decisions.
Deployment readiness	Not production-ready	Final pipeline compatibility, monitoring and governance remain necessary.

Business suitability

The final model is suitable as an advisory decision-support and risk-prioritisation component for the proposed HealthConnect solution. It can help administrative teams identify appointments carrying an elevated no-show risk, allowing finite outreach resources to be focused on those appointments that could benefit most from additional support. Given its moderate predictive discrimination and notable false-positive/false-negative rates, model predictions should always complement — rather than replace — professional administrative judgement.

Appropriate uses

- Prioritising supportive, proactive patient outreach and scheduling reminders.
- Streamlining and prioritising administrative workflows for clinic coordinators.
- Pinpointing specific appointments that could benefit from manual reconfirmation efforts.
- Monitoring overall clinic-level or cohort-level non-attendance trends.
- Assessing the operational effectiveness of targeted patient-attendance support programmes.

Non-intended uses

- Denying, delaying or restricting patient access to healthcare services.
- Automatically cancelling or rescheduling appointments based on risk-score thresholds.
- Imposing administrative or financial penalties on patients flagged as high risk.
- Making clinical, triage, diagnostic or treatment decisions.
- Assuming that predictive model features are direct, causal reasons for individual patient non-attendance.
- Substituting automated flags for human empathy, discretion and administrative judgment.

Operational status

The final model is ready for integration testing and demonstration; this status does not represent production readiness. Transitioning to production would require formal pipeline-compatibility verification, stakeholder-approved operational thresholds, reliable patient-identifier mapping (traceability), and post-deployment performance and drift monitoring.

9. Limitations, risks and dependencies

The final model should be interpreted within the limitations identified throughout Weeks 5–8. These items do not invalidate the modelling work; they define the conditions under which the model should be interpreted, integrated and monitored. A full review of assumptions,

limitations, risks and dependencies across all tracks and weeks is provided as a standalone companion document: the Week 7 HealthConnect Assumptions, Limitations, Risks & Dependencies Report. The register below is specific to the final Week 8 model.

Risk or limitation	Evidence / impact	Week 8 treatment	Status
Moderate predictive discrimination	Final holdout ROC-AUC is approximately 0.689.	Present model as moderate risk prioritisation rather than highly accurate prediction.	Documented
False-negative risk	136 observed no-shows were not flagged at the 0.45 holdout threshold.	Retain recall as an important monitoring metric.	Documented
False-positive outreach burden	225 attending appointments were flagged at the 0.45 holdout threshold.	Monitor outreach volume and false-positive burden.	Documented
Operational threshold not formally approved	0.45 improves recall but increases supportive-outreach requirements.	Keep 0.45 provisional pending stakeholder operational agreement.	Open
Patient-grouped CV unavailable	Processed modelling matrices did not support patient-grouped cross-validation.	Document and recommend patient-grouped validation when identifiers are available.	Documented
Preprocessing not fully nested within CV	Existing processed matrices were used rather than rebuilding preprocessing inside each fold.	Recommend fully nested preprocessing in future validation.	Documented
Segment-level performance variation	Week 7 identified different FP/FN behaviour across appointment segments.	Retain segment-level monitoring.	Monitor
Prediction-level identifier traceability	Prediction-level linkage requires stronger appointment-identifier preservation.	Resolve identifier traceability during engineering integration.	Open
Cross-track data-version / missing-distance discrepancy	A previously identified discrepancy in missing-distance counts remains a data-governance dependency.	Carry forward as a documented data-quality/governance issue.	Open
Observational rather than causal evidence	Analytics and modelling relationships should not be interpreted as causal effects.	Use association language and avoid causal claims.	Documented
Potential future data drift	Future appointment behaviour may differ from historical development	Recommend post-integration drift/performance	Future monitoring

Risk or limitation	Evidence / impact	Week 8 treatment	Status
	data.	monitoring.	
ML Engineering compatibility	Final end-to-end pipeline reproduction remains pending until HC_W8_TEST_001 is returned.	Final model, schema and reproducible reference test supplied to ML Engineering; end-to-end reproduction remains a post-submission dependency.	Open — handoff completed; return validation unavailable before submission

10. Cross-track contribution

Data Analytics → Data Science

During Week 7, Data Analytics findings were compared directly against patterns reproduced within the Data Science modelling sample. An initial difference in lead-time category definitions was identified and resolved by harmonising the Data Science retest boundaries with the Analytics definitions before executing the final retests. Week 8 carries this validated cross-track evidence base forward rather than repeating the Week 7 analysis steps.

Cross-track test	Expected from Analytics	Retest result
Lead-time direction	No-show rate increases with longer lead time	PASS
Previous-no-show-history direction	No-show rate increases with prior no-show history	PASS
Definition alignment	Use 0–7, 8–21, 22–40, 41–60 lead-time bands	PASS — Data Science retest now uses Analytics bands

Both directional findings — rising no-show prevalence with longer booking lead time, and with greater previous no-show history — were reproduced within the Data Science modelling sample after harmonising the lead-time definitions, confirming strong cross-track consistency. These validated findings were incorporated directly into the final Data Science contribution: they contextualise the Week 7 lead-time and history segment error patterns, support the interpretation of the corresponding engineered interaction features (`history_x_lead_time`, `distance_x_lead_time`, `history_interaction`), and underpin the business interpretation in Section 8. This alignment remains strictly observational, however, and must not be read as evidence that booking lead time or prior no-show history causes non-attendance. Week 8 integration outcome: PASS — the validated Analytics findings are incorporated into the final model interpretation and HealthConnect business narrative.

Data Science → ML Engineering

The Data Science → ML Engineering handoff communicates the final tested model and its technical requirements. The handoff preserves the exact final serialised model artefact, the 38-feature input contract with exact feature names and ordering, preprocessing and scaling expectations, probability-output semantics (predicted probability of No-Show), the provisional 0.45 decision threshold, explicitly documented model limitations, and a reproducible single-row integration reference test (HC_W8_TEST_001). Communication alone is not treated as

sufficient: successful integration requires verifying that the target engineering pipeline exactly reproduces the Data Science model's output probabilities and flag outcomes on the provided test case.

Field	Specification
Test ID	HC_W8_TEST_001
Objective	Confirm that the ML Engineering scoring pipeline reproduces the final Data Science model output.
Expected no-show probability	0.4951596731
Provisional threshold	0.45
Expected risk flag	1 (Higher No-Show Risk)
Probability tolerance	1e-6
Pass condition	Absolute probability difference $\leq 1e-6$ AND identical risk flag
If test fails	Check model version, feature order, preprocessing, positive-class probability selection and threshold implementation; correct the identified issue and rerun the same test.

The Data Science → ML Engineering handoff package was fully completed and dispatched prior to the submission deadline, including the final serialised Logistic Regression model, the 38-feature input contract, extensive model metadata, and the official single-row reference input for verification. However, no pipeline result was returned by ML Engineering before the final deadline. End-to-end engineering reproduction therefore remains an outstanding integration dependency and is not reported as validated; this is an outstanding operational dependency for final deployment testing, not an issue with the underlying predictive model or the validity of the handoff itself.

Final cross-track integration record

Integration	Output exchanged	Final status
Data Analytics → Data Science	Lead-time and previous-no-show-history findings	PASS
Data Science → ML Engineering	Final model, 38-feature contract, metadata and reference test case	HANDOFF COMPLETED
ML Engineering → Data Science	No return output received before Data Science submission	PENDING — NO RESPONSE RECEIVED

11. Final model documentation

Model handoff contract

Requirement	Specification
Model artefact	healthconnect_week7_final_model.pkl
Model type	LogisticRegression

Requirement	Specification
Final feature count	38
Feature schema	healthconnect_week7_final_feature_schema.csv
Prediction output	Predicted probability
Positive class	No-Show
Provisional threshold	0.45
Risk-flag rule	risk_flag = no_show_probability >= 0.45
Threshold status	Provisional — requires operational approval
Required integration check	ML pipeline probability must reproduce DS probability within agreed tolerance

Compatibility checks confirm that the model and schema are internally consistent: the serialised model expects exactly 38 input features, matching the finalised schema (38 = 38, Pass), the schema contains 38 unique feature names (Pass), and the final feature order is frozen in the published ML feature contract. The processed Week 7 holdout matrix used for pipeline verification (966 rows) was independently confirmed to contain the expected 38 features in the exact schema order (PASS). All pre-handoff validation checks passed, confirming the component was fully prepared for pipeline compatibility and reproduction testing by ML Engineering.

Final model card

Model card item	Value
Model name	HealthConnect Final No-Show Risk Model
Model family	Logistic Regression
Business objective	Support prioritisation of appointments at elevated no-show risk
Prediction target	Appointment attendance outcome
Positive class	No-Show
Final feature count	38
Output	Predicted probability of No-Show
Provisional threshold	0.45
Holdout ROC-AUC	0.6890
Holdout PR-AUC	0.6717
Holdout precision	0.6066
Holdout recall	0.7184
Holdout F1	0.6578
False negatives	136
False positives	225

Model card item	Value
Primary intended use	Supportive administrative risk prioritisation
Non-intended use	Clinical decisions, access restriction, automatic cancellation, patient penalties, or causal inference
Week 7 refinement decision	Calendar-feature ablation rejected under predefined recall-retention criterion
Cross-track validation	Analytics directional validation PASS
Analytics findings incorporated	Validated lead-time and previous-no-show-history patterns used for model, segment and business interpretation
Integration status	DS → MLE handoff completed; MLE → DS pipeline reproduction not independently verified before submission because no return result was received
Production status	Integration/demo candidate — not production-ready

The final model and associated handoff artefacts — the ML feature contract, model metadata, model card, reference input, and expected-output specification for HC_W8_TEST_001 — were saved alongside the Week 8 notebook as the formal, reproducible record of the final Data Science component.

12. Mandatory HC-POD final integration

Per the Week 8 brief, this section documents the final connection between the Data Science component and each other HealthConnect track, in the mandatory nine-field format.

Data Analytics → Data Science

1. Track collaborated with: Data Analytics.
2. Dependency: validated attendance-pattern findings (booking lead time, previous no-show history).
3. Output received: lead-time and previous-no-show-history directional findings and recommended lead-time bands.
4. Output provided: harmonised retest results confirming the findings within the Data Science modelling sample.
5. Final integration activity: Data Science retested both Analytics findings against the modelling sample, after harmonising lead-time band definitions to match Analytics' boundaries.
6. What changed: the Data Science lead-time retest now uses the Analytics-defined 0–7, 8–21, 22–40, 41–60 bands rather than the earlier internal definition.
7. How the change improved the solution: both directional findings were reproduced, giving a validated, cross-track evidence base for interpreting the model's lead-time and history-related features and their error patterns.
8. Evidence: cross-track validation summary table and harmonised lead-time/history retest result files (all retests PASS).

9. Contribution to final presentation: supports the model-interpretation and business-interpretation narrative, demonstrating that Data Science conclusions are independently corroborated by Data Analytics.

Data Science → ML Engineering

1. Track collaborated with: ML Engineering.
2. Dependency: ML Engineering requires the final model artefact and its exact technical scoring contract to integrate it into the final workflow.
3. Output received: none prior to submission (see ML Engineering → Data Science below).
4. Output provided: the final serialised Logistic Regression model, the 38-feature input contract with frozen ordering, preprocessing/scaling expectations, probability-output semantics, the provisional 0.45 threshold, documented limitations, and the HC_W8_TEST_001 reference input and expected output.
5. Final integration activity: a formal handoff package and integration test specification were prepared and dispatched, requesting that ML Engineering score the reference case and return the resulting probability, risk flag, feature-acceptance confirmation and any compatibility issues.
6. What changed: the final model moved from an internally-validated Data Science artefact to a packaged, versioned handoff ("HealthConnect_Week8_Final_LR") with a reproducible pass/fail integration test.
7. How the change improved the solution: the model is now accompanied by an unambiguous, testable scoring contract (exact feature order, probability tolerance of 1e-6, and explicit pass condition), rather than an informal description, reducing the risk of silent integration errors.
8. Evidence: ML feature contract, model metadata file, model card, reference input/output files, and the dispatched integration-test specification for HC_W8_TEST_001.
9. Contribution to final presentation: demonstrates a complete, engineering-ready model handoff and defines precisely what "successful integration" means for the final HC-POD walkthrough.

ML Engineering → Data Science

1. Track collaborated with: ML Engineering.
2. Dependency: Data Science requires confirmation that the ML pipeline reproduces the model's probability and risk-flag output within tolerance.
3. Output received: no pipeline result for HC_W8_TEST_001 was returned before the Data Science submission deadline.
4. Output provided: not applicable — this dependency runs from ML Engineering to Data Science.
5. Final integration activity: a return-validation procedure was prepared in the Week 8 notebook to compare the ML Engineering pipeline probability against the frozen Data Science reference output as soon as it is received.
6. What changed: nothing yet — the comparison logic is implemented and ready, but has not been executed against a real pipeline result.

7. How the change improved the solution: not applicable at submission time; this activity is deliberately reported as pending rather than claimed as validated, consistent with the project's standard of not reporting an unsupported pass.
8. Evidence: the prepared HC_W8_TEST_001 comparison cell (status: PENDING) in the Week 8 notebook, and the integration record in Section 10.
9. Contribution to final presentation: identifies the single outstanding cross-track dependency to be resolved before the Data Science component can be reported as fully production-integrated.

13. Presentation materials

13.1 Model section of the final presentation (technical)

For the technical portion of the final HC-POD presentation, the Data Science contribution should be presented as follows: the final model is a 38-feature refined Logistic Regression that estimates the individual probability of appointment no-show, evaluated on a held-out test set with ROC-AUC 0.689, PR-AUC 0.672, precision 0.607, recall 0.718 and F1 0.658 at the provisional 0.45 threshold (347 true positives, 258 true negatives, 136 false negatives, 225 false positives). Fold-level cross-validation against the Week 5 baseline showed mixed results (ROC-AUC improved in 3 of 5 folds, PR-AUC in 2 of 5), so the model should be framed as a moderate, interpretable risk-prioritisation tool rather than a major predictive breakthrough. The Week 7 calendar-feature ablation was tested and rejected under a predefined 0.002 recall-retention tolerance, so the full 38-feature specification was retained. The final model, its feature contract, metadata and a reproducible reference test case (HC_W8_TEST_001) have been handed off to ML Engineering; return-pipeline validation remains the one outstanding integration dependency.

13.2 Concise model summary for non-technical stakeholders

In plain terms: the Data Science team built a tool that estimates, for each upcoming appointment, how likely a patient is to miss it. The goal is not to punish or restrict patients, but to help clinic staff focus limited reminder and outreach effort on the appointments most likely to be missed. In testing, the tool correctly flagged about 7 in 10 appointments that were actually missed, but it also flagged a meaningful number of appointments that patients did in fact attend — so every flag should be treated as a prompt for a supportive check-in, not an automatic decision. The tool is ready to be connected into HealthConnect's systems for a trial, but it is not yet a finished, production system: the clinic still needs to agree on how sensitive the tool should be, and the engineering team still needs to confirm it reproduces the Data Science team's results exactly once it is wired into the full pipeline.

13.3 Executive summary

- **The challenge:** missed appointments create avoidable pressure on clinic capacity and reduce opportunities to support patients before scheduled visits.
- **The solution:** the Data Science team developed a predictive model that assigns each appointment a probability of non-attendance, designed to help administrative teams prioritise supportive actions (reminders/reconfirmations) rather than automate patient decisions.

- **Operational performance:** on holdout data, the model identifies a meaningful proportion of no-shows but also flags some patients who will attend; it serves best as an advisory decision-support tool rather than an automated driver of high-impact policies.
- **Cross-track integration:** independent attendance patterns discovered by Data Analytics (higher no-show rates with longer booking lead times and previous no-show history) were retested, verified and aligned with the Data Science sample, providing critical real-world context for model interpretation and segment behaviour.
- **Engineering handoff:** the final model was packaged with a fixed 38-feature input contract and a reproducible reference test for ML Engineering. The Data Science handoff was completed, but no pipeline result was returned before submission, so end-to-end engineering reproduction remains an outstanding integration dependency and is not reported as validated.
- **Conclusion:** the resulting component delivers an interpretable, tested, and cross-track-aligned risk-prioritisation system, while documenting its operational boundaries, trade-offs, and rules for responsible use.

A full individual video presentation (5–10 minutes), covering introduction, track contribution, cross-track collaboration, testing/refinement outcomes, overall solution fit, and closing reflection, is prepared as a separate deliverable in line with the Week 8 brief's individual-submission requirement, and is not reproduced in this written report.

14. Week 8 readiness and submission status

Readiness item	Status
Final candidate model confirmed	Complete
Final 38-feature specification frozen	Complete
Final evaluation consolidated	Complete
Baseline comparison documented	Complete
Refinement decision documented	Complete
Error-analysis implications documented	Complete
Business suitability documented	Complete
Intended/non-intended use documented	Complete
Analytics → DS validation incorporated	Complete
DS → ML Engineering handoff completed	Complete
ML Engineering → DS pipeline reproduction	Pending — no response
Limitations and risks documented	Complete
Final model card prepared	Complete
Executive summary prepared	Complete
Presentation/video	Separate deliverable

The HealthConnect Data Science component is considered complete for Week 8 submission, with one explicitly flagged outstanding dependency: ML Engineering did not return the

pipeline result for HC_W8_TEST_001 before submission, so ML Engineering → Data Science pipeline reproduction remains pending and is not reported as validated. The final model is considered suitable for HealthConnect integration and demonstration as a supportive risk-prioritisation component. It is not considered production-ready: the 0.45 decision threshold remains provisional, and future deployment would require confirmed pipeline compatibility, operational-threshold approval, identifier traceability, monitoring and governance, and continued validation.

Data Science contribution to the end-to-end HealthConnect solution

The final Data Science component does not operate in isolation. Appointment data is collected at scheduling; Data Analytics validates broad population-level attendance patterns and historical business context; Data Science generates individual, appointment-level no-show probabilities using the predictive model; ML Engineering deploys the predictive scoring component into a reproducible production pipeline; administrative teams use the resulting risk scores to target supportive outreach workflows; and clinic staff retain full agency and accountability over operational scheduling decisions. Data Science specifically contributes the final trained 38-feature Logistic Regression model, the exact technical contract (feature schema, preprocessing expectations and column order) needed for scoring, individualised risk-probability outputs rather than raw binary classifications, empirical operational-threshold guidance, exhaustive validation records, and explicit responsible-use boundaries — collectively enabling HealthConnect to target finite outreach resources toward appointments carrying elevated non-attendance risk through a structured, evidence-based, non-automated workflow.

References

- AnalystLab Africa. (2026a). *HealthConnect Clinic: Baseline modelling report*. Week 5 submission.
- AnalystLab Africa. (2026b). *HealthConnect Clinic: Model improvement & validation report*. Week 6 submission.
- AnalystLab Africa. (2026c). *HealthConnect Clinic: Model testing & refinement report*. Week 7 submission.
- AnalystLab Africa. (2026d). *HealthConnect Clinic: Assumptions, limitations, risks & dependencies report*. Week 7 submission.
- AnalystLab Africa. (2026e). *HealthConnect Clinic: Data Analytics track Week 7 testing and validation report*. Week 7 submission.
- AnalystLab Africa. (2026f). *AnalystLab Africa Week 8 Data Analytics Assignment: HealthConnect final integration, presentation & project showcase*. Week 8 assignment brief.
- AnalystLab Africa. (2026g). *HealthConnect Clinic project plan*. Week 4 submission.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide)* (7th ed.). Project Management Institute.
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>